

3-level dataset with level-1 time-series

How should I analyze it?

Luca Menghini^{1,2}

¹Department of General Psychology, University of Padova, Italy

²HIT Research Centre, University of Padova, Italy



Psicostat - The Royal Rumble

Padova, October 30th 2025

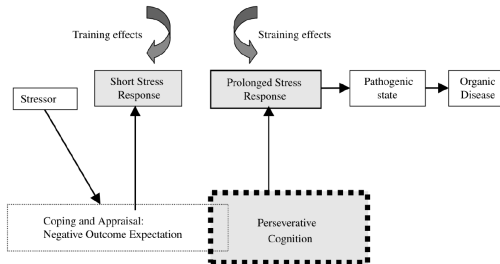


Stress responses & perseverative cognitions

Perseverative cognitions (PC) = “repeated or chronic activation of the cognitive representation of one or more psychological stressors”

(Brosschot et al., 2006)

- Re-enactment of past stressors (**rumination**) and anticipation of future stressors (**worry**)
- PC trigger and prolong stressor-related physiological activation, possibly leading to long-term health consequences
- Main pathogenic mechanism of psychological stress?

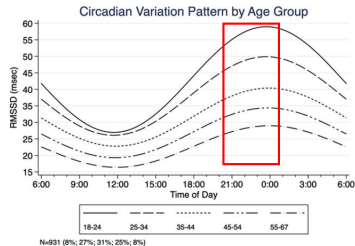
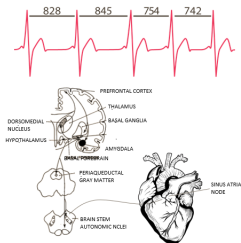


Sleep quality and primary insomnia: The hyperarousal hypothesis

Sleep-related Heart Rate Variability

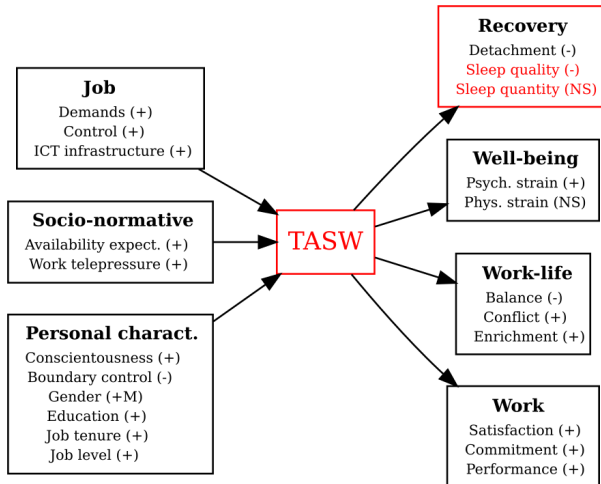
(Resting-state) HRV = Beat-to-beat fluctuations in inter-beat intervals in phase with respiratory cycles

- Index of cardiovascular health & vagal cardiac control (Shaffer et al, 2014; EU Society of Cardiology, 1996)
- Primary allostatic load mediator, primary output of Central Autonomic Network, predictive of individual differences in emotion regulation, sensitive to stress and workplace stress (Jarczok et al, 2013)
- Increasing trend throughout the night



Technology-Assisted Supplemental Work (TASW)

= Performance of “role-prescribed tasks at home after regular work hours with the aid of technological tools”
(Fenner & Renn, 2010)



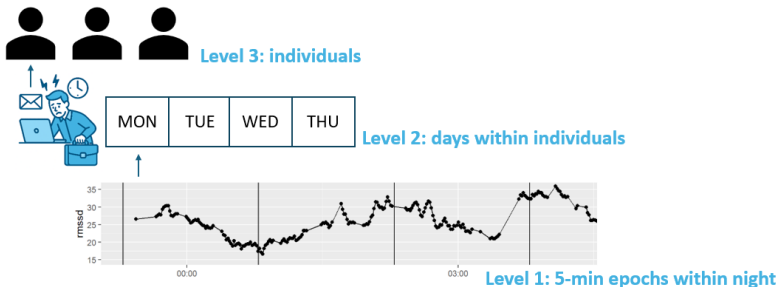
Kühner et al. (2023)

J Vocat Behav

$K = 89$ indep. samples,

$N = 39,085$ employees

Predicting sleep-related HRV trajectories by day-level TASW and PC within individuals



Background and aim
○○○○○

Method
●○○

How should I analyze it?
○○○○○○

Discussion
○○

DARE project

The sample so far



52 employed adults (46.2%F, 41.4 ± 11.7 y)

Job tenure: 12.8 ± 10.4 y, 44.2 ± 6.4 hrs/week

75% employees, 4% managers, 21% other

62% remote/hybrid workers

Recruitment

Convenience sampling, 5-day EMA $\rightarrow$ €10-25 reward + individual report

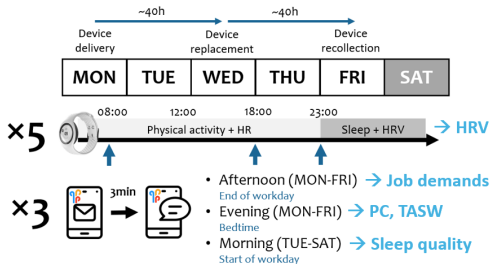
756 observations, 260 workdays

Response rate = $96.9\% \pm 5\%$

Exclusion criteria:

- Unemployed/Students/Trainees
- < 6 work hours/day
- Nocturnal shifts
- Careless resp. (attention checks)
- Manual workers
- Full remote workers

Ecological Momentary Assessment



RMSSD (ms) Minute-by-minute overlapping 5-min epochs with a lot of missing data! :(

TASW (1-7) (Fenner & Renn 2010)
4 items, back-translated (e.g. “I was available for colleagues, customers, and boss until I went to bed”) $\omega_{within} = 0.75$, $\omega_{between} = 0.92$

Mini Sleep Qs (Natale et al. 2008)

4 items from “sleep” subscale

e.g., *Difficulties in falling asleep* $\omega_{within} = 0.87$, $\omega_{between} = 0.99$

Mini Sleep Qs (1-7) (Natale et al. 2008)

4 items from “sleep” subscale

e.g., *Difficulties in falling asleep* $\omega_{within} = 0.7$, $\omega_{between} = 0.71$

Alternative statistical methods for modeling 3-level time series

- Night-specific slope as Y variable
- 3-level LMER
- Growth curve model
- Dynamic SEM

Night-specific slope as Y variable: Back to 2-level comfort zone

```
# step 1
for(id in unique(ema$ID)){
  for(night in 1:4){
    ema[ema$ID==id & ema$day==night,"HRV_slope"] <-
      summary(
        lm(RMSSD ~ time + RMSSD_lag, data = sleepHRV)
      )$coefficients[2,1]
  }
}

# step 2
m <- lmer(HRV_slope ~ pc + tasw + (1|ID), data = ema)
```

3-level LMER

Will it ever converge?

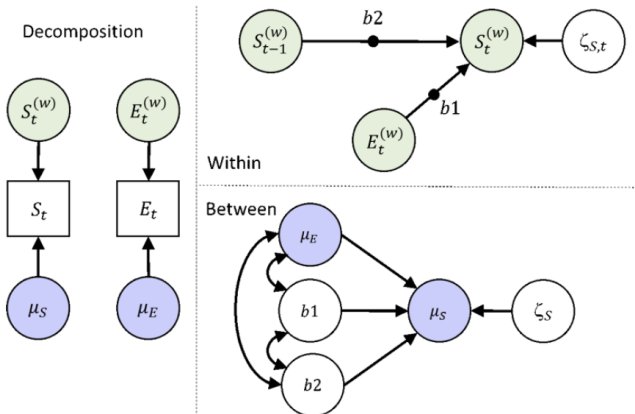
```
m <- lmer(RMSSD ~ time + time:pc + time:tasw +  
          RMSSD_lag + (1|ID:day),  
          data = sleepHRV)
```

Growth curve model with MPLUS

```
VARIABLE: NAMES ARE ...;  
          USEVAR ARE ...;  
          WITHIN = time;  
!#and what about day-level variables?  
  
ANALYSIS: TYPE = THREELEVEL RANDOM;  
          ESTIMATOR = MLR;  
  
MODEL: %WITHIN%  
       slope | RMSSD ON time;  
  
       %BETWEEN%  
       slope ON tasw;  
       slope ON pc;
```

Not sure about how doing this with lavaan...

Dynamic SEM (???)



Idiographic latent profile analysis (???)



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Idiographic Momentary Profiles of Personality Facets

Colin J. Lee and Emorie D. Beck

Department of Psychology, University of California, Davis

“The present study used idiographic (i.e., $N = 1$) LPA to extract and group together similar measurement occasions from an individual’s time series data. Broadly, LPA uses continuous (or ordinal; Van Lissa et al., 2024) indicators to identify subgroups in the data. In our case, LPA probabilistically classifies each measurement occasion into a subgroup, separately for each person. The subgrouping procedure results in a set of similar moments that can be summarized as a personality facet profile (15 total variables in each profile) unique to the individual.⁶ A measurement occasion is grouped with another if both have sufficiently similar indicator values and covariances. Idiographic LPAs were estimated using the Mclust package (Scrucca et al., 2016, 2023) in R (R Core Team, 2024).”

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Discussion?

Thank you!

PSICOSTAT ROYAL RUMBLE

Padova, October 30 2025

luca.menghini@unipd.it

 [profile/Luca-Menghini](#)  [@LuMenPsy](#)  [in/menghiniluca](#)  [Luca-Menghini](#)

R code available from <https://luca-menghini.github.io/pResentations>